FemboyEngine 0.2.0-alpha.1 — getting started

This document lists what to install and which files must sit next to the engine to run the editor, the runtime and the development tools on Windows, Linux and macOS, plus the toolchain needed for Android and iOS builds.

Supported platforms

Platform	Editor	Game runtime	Toolchain
Windows 10/11 x64	yes	yes	clang-cl (primary), GCC 16+ (second opinion)
Linux x64	yes	yes	Clang 19+, GCC 16+
macOS 13+ (Apple Silicon and Intel)	yes	yes	Apple Clang from Xcode, Metal backend
Android 5.0+ (API 21)	no	yes, through SDL3	NDK r28c, Android Gradle Plugin 8.7.3, Gradle 8.12
iOS 11+	no	yes, through SDL3	Xcode 16+, iOS SDK 14.2+

The editor is a desktop application. Games build for every platform listed above; mobile targets use the SDL3 platform layer.

Required software

MSVC (the `cl.exe` compiler) is not required. On Windows the project builds with clang-cl from LLVM; the MSVC compiler is never invoked. Visual Studio Build Tools are needed only as the supplier of the Windows SDK and the standard library headers that clang-cl uses.

Tool	Minimum	Validated	Purpose
CMake	3.28	3.31.6	configure and build
Ninja	1.11	1.12.1	build generator
LLVM / clang-cl	19	23.1.4	primary Windows compiler
Clang (Linux/macOS)	19 / Xcode 16	—	Linux and macOS compiler
GCC	15	16.2.0	second compiler for portability checks
Python	3.11	3.13	documentation and notices generation
Git	2.40	—	clone and version control
Vulkan Loader	1.4	1.4	runtime on Windows and Linux
MoltenVK	1.2	—	Vulkan over Metal on macOS (or the Metal backend directly)
7-Zip	21	25.01	unpack the LZMA SDK (otherwise the built-in FbyLZ is used)
Android Studio	—	NDK r28c	Android SDK, NDK

			28.2.13676358, compileSdk 35
OpenJDK	17	17	Gradle packaging of the Android build
Xcode	16	—	iOS and macOS builds

The language standard is C++26, so an older compiler will not work even when its CMake version matches.

Quick start with the setup scripts

The scripts live in `SetupScripts/` and are split by software group. The one-click scripts install everything; the component scripts install one group.

Group	Windows	macOS	Linux
C++ toolchain, CMake and source dependencies	`SetupScripts\SetupCpp.bat`	`./SetupScripts/SetupCpp.command`	`./SetupScripts/SetupCpp.sh`
7-Zip archiver	`SetupScripts\SetupArchiver.bat`	`./SetupScripts/SetupArchiver.command`	`./SetupScripts/SetupArchiver.sh`
Optional DLSS/FidelityFX/XeSS SDKs	`SetupScripts\SetupUpscalers.bat`	`./SetupScripts/SetupUpscalers.command`	`./SetupScripts/SetupUpscalers.sh`

The all-in-one entry points remain `Setup.bat`, `Setup.command` and `Setup.sh`. The C++ scripts install the compiler, CMake, Ninja, the Vulkan headers and the pinned source dependencies. On Windows, `SetupCpp` installs the official LLVM 23.1.4 MSI and verifies the `clang-cl` version; the Unix scripts install the distribution Clang (19+). The archiver scripts install 7-Zip. The upscaler scripts only verify or download vendor SDKs; they are optional and never part of the engine repository.

Useful flags: `--skip-system` leaves system packages alone (a CI image already has the tools); `--force` re-downloads every C++ archive. The C++ script writes `third_party/dependencies.lock` with the pinned versions and archive hashes. At the end it prints the mobile toolchain report: NDK, JDK and Xcode.

From a GitHub archive

1. Open the repository page and choose **Code > Download ZIP**.
2. Extract the archive to a directory without spaces or non-ASCII characters.

3. Run `SetupCpp` and `SetupArchiver` for the host platform.
4. Optionally run `SetupUpscalers` and enable a vendor backend.
5. Configure and build:

```
cmake --preset dev-clang
cmake --build --preset dev-clang
ctest --preset dev-clang --output-on-failure
```

The archive contains no vendor SDK. The C++ script installs the toolchain and materialises the pinned sources; the archiver script installs 7-Zip. The upscaler scripts are only needed when a DLSS/FidelityFX backend is explicitly enabled. Without them, FSR1, the FSR2-compatible core path and custom frame generation build and run with runtime fallback.

Manual installation

Without the script the steps are the same: install a compiler, CMake, Ninja, the Vulkan headers, and lay out the third-party sources.

Windows

6. Install LLVM 23.1.4 (clang-cl) and add its `bin` directory to `PATH`.
7. Install Visual Studio Build Tools with the "C++ Build Tools" workload (Windows SDK and standard library headers). The MSVC compiler itself is not used.
8. Install CMake 3.31 and Ninja 1.12 (the Build Tools copy is fine).
9. Install the Vulkan SDK 1.4 — it provides `vulkan-1.dll` and `VULKAN_SDK`.
10. Install 7-Zip (optional).
11. Download the archives the script would fetch and unpack them into the same `third_party/` directories: SDL3 `release-3.4.18`, Vulkan-Headers `v1.4.366`, glslang `15.3.0`, miniaudio `0.11.23`, Box2D `v3.2.0`, RNNoise `0.1.1`, Monocypher `4.0.2`. HarfBuzz and SheenBidi need no download: they are already in the repository.

Linux

```
sudo apt install build-essential clang cmake ninja-build git pkg-config libvulkan-dev vulkan-tools p7zip-full
```

Then lay out the same archives as on Windows. GCC 16 comes from the distribution or a separate build; Clang is enough for the primary build.

macOS

```
xcode-select --install
brew install cmake ninja git p7zip glslang vulkan-headers molten-vk llvm
```

Xcode Command Line Tools are mandatory: they provide `clang++`, the Metal frameworks and the SDK headers. The Metal backend needs no Vulkan SDK.

Building

From the repository root:

```
cmake -S . -B build-dev -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo -DFBY_BUILD_TESTS=ON
cmake --build build-dev
ctest --test-dir build-dev --output-on-failure
```

On Windows the same commands run inside a `vcvars64` environment; `SetupScripts/SetupWindows.ps1` prepares the dependencies and the toolchain. Main CMake options: `FBY_BUILD_EDITOR`, `FBY_BUILD_TESTS`, `FBY_BUILD_TOOLS`, `FBY_BUILD_SANDBOX`, `FBY_BUILD_SERVER`, `FBY_BUILD_LOBBY`, `FBY_WITH_ASAN`, `FBY_RHI_BACKEND` (`auto`, `vulkan`, `metal`, `null`).

The executables are grouped by purpose:

Directory	Contents
`build-dev/bin`	applications: `fby_editor`, `fby_sandbox`, `fby_server`, `fby_lobby`, platform plugins and `plugins/`
`build-dev/bin/tools`	console tools: `fby_fontc`, `fby_textc`, `fby_fempak`, `fby_scene_tool`, `fby_template_gen`, `fby_aotc`, `fby_loc_tool`, `fby_shaderc`
`build-dev/bin/tests`	test executables (`fby_*_test`) that `ctest` runs
`build-dev/bin/lib`	static libraries
`build-dev/bin/plugins`	native engine plugins

The `*.pdb` files next to the executables are debug symbols written by the compiler. They are only used by a debugger and are not needed to run anything; a release package does not have to ship them.

Code editor and IDE

The engine does not ship its own text editor. Development uses an external editor that understands the CMake project and `compile_commands.json`.

Recommended setup:

- **VS Code**;
- the **clangd** extension;
- the **CMake Tools** extension;
- **CodeLLDB** or `cppdbg` for debugging;
- RenderDoc for GPU debugging.

CLion also works: it understands CMake natively and can launch `fby_editor` and the tests. Visual Studio can be used as an IDE, but not as an MSVC build; see below.

VS Code setup

12. Install VS Code.
13. Install the extensions:

- `llvm-vs-code-extensions.vscode-clangd`;
- `ms-vscode.cmake-tools`;
- `vadimcn.vscode-lldb` or `ms-vscode.cpptools` for `cppdbg`.

14. Open the repository root (the folder that contains `CMakePresets.json`).
15. Disable the built-in IntelliSense; otherwise it competes with clangd:

```
{
  "C_Cpp.intelliSenseEngine": "disabled"
}
```

16. Select the `dev-clang` or `dev-clangcl` CMake preset.
17. Configure:

```
cmake --preset dev-clang
```

18. Build:

```
cmake --build --preset dev-clang
```

19. Run tests:

```
ctest --preset dev-clang
```

The repository already contains:

- `CMakePresets.json` — the presets;
- `.clangd` — compile-database path;
- `.vscode/settings.json` — clangd and CMake Tools settings;
- `.vscode/launch.json` — editor and `frame_composer` launch targets.

clangd

clangd must use the same Clang that builds the project:

```
{
  "clangd.arguments": [
    "--compile-commands-dir=${workspaceFolder}/build-audit",
    "--query-driver=C:/Users/<user>/AppData/Local/Programs/LLVM/bin/clang++.exe"
  ]
}
```

`--query-driver` lets clangd find the Windows SDK headers. If a different build directory is used, change `--compile-commands-dir` or `.clangd`.

Do not enable C/C++ IntelliSense and clangd at the same time: that produces two different diagnostic and navigation stacks.

Build presets

Preset	Compiler	Purpose
`dev-clang`	`clang++`	Debug, tests, `compile_commands.json`
`release-clang`	`clang++`	RelWithDebInfo, tests
`dev-clangcl`	`clang-cl`	Debug with the validated Windows toolchain
`audit-clang`	`clang++`	existing `build-audit` for clangd

Different compilers must use **different build directories**. Do not reconfigure `build-audit` from clang++ to clang-cl: the CMakeCache belongs to the old toolchain.

Debugging

For `cppdbg`, use gdb from MinGW/LLVM; CodeLLDB uses LLVM's lldb. Example:

```
{
  "name": "Editor (Debug)",
  "type": "cppdbg",
  "request": "launch",
  "program": "${workspaceFolder}/build-dev/bin/fby_editor.exe",
  "cwd": "${workspaceFolder}",
  "MIMode": "gdb",
  "miDebuggerPath": "gdb.exe"
}
```

Use RenderDoc for GPU debugging: attach it to `fby_editor.exe` and capture a frame without embedding a debugger in the editor.

MSVC support

No. The project is deliberately built around Clang:

- C++26 and the strict flag set are validated with Clang 23.1.4;
- clang-cl already provides the Windows ABI, Windows SDK and an MSVC-compatible driver;
- the MSVC compiler (`cl.exe`) is not required;
- adding MSVC means a second set of flags, warnings, STL behavior and C++26 feature support, plus a separate CI matrix.

That does not mean MSVC is forbidden forever. If it is needed as an **optional second compiler**, do it as a separate task:

20. add a `dev-msvc` preset;
21. verify the C++26 features used by the engine;
22. add `FBY_COMPILER_MSVC` for flag differences;
23. add a separate CI job;
24. keep Clang as the primary toolchain, not a replacement.

Until then, the primary and validated path is **Clang / clang-cl**. Visual Studio may be used as an IDE with the clang-cl toolchain, but not as an MSVC build replacement.

Running the editor

The editor opens the project from its **current working directory**: it looks for `project.femproj`, mounts `assets/` as the virtual file system, writes project editor settings to `editor.ini` next to `project.femproj`, and keeps the ImGui layout in `Saved/Config/editor.ini`. Start it from the project root (the repository root for the demo project).

```
cd %USERPROFILE%\FemboyEngine
build-dev\bin\fby_editor.exe
```

Linux and macOS:

```
cd /path/to/project
./build-dev/bin/fby_editor
```

Everything the engine writes at runtime goes to `Saved/` in the working directory, so the project directory stays clean:

Path	Contents
`Saved/Logs/<category>/<category>-YYYYMMDD-HHMMSS.log`	the full log of a run: `engine`, `editor`, `sandbox`, `server`, `lobby`. The 20 newest files per category are kept and older ones are removed automatically
`Saved/Crashes/<YYYYMMDD-HHMMSS>-<pid>/report.txt`	crash report: time, application, pid, reason, address and, on Linux and macOS, a backtrace
`Saved/Crashes/<...>/crash.dmp`	Windows minidump for a debugger
`Saved/Config/editor.ini`	editor settings and panel layout
`Saved/Cache/`	cache of intermediate data

Environment variables: `FBY_STORAGE_DIR` moves the `Saved/` root (for example to a disk with space) and `FBY_LOG_FILE=0` disables the log file while the console keeps working.

Files that must be present

Path	Required	Purpose
`build-dev/bin/fby_editor`	yes	the editor itself
`assets/`	yes	fonts, localization, scenes, atlases, project configs
`assets/fonts/IBMPlexSans-Regular.ttf`, `...-Bold.ttf`, `...-Italic.ttf`, `...-BoldItalic.ttf`, `IBMPlexMono-Regular.ttf`	yes	the font family used by the editor and by scene text
`assets/fonts/IBMPlex.family.toml`	yes	family manifest with the five style slots
`assets/fonts/Inter.fbyfont`	yes	pre-baked atlas used by the legacy UI path
`assets/loc/ui.fbin`	yes	interface localization table
`assets/scenes/*.toml`	yes	project scenes
`project.femproj`	yes	project description: default scene, simulation, render, accessibility
`build-dev/`	yes	the CMake build tree: executables and build results live there
`build-dev/bin/fby_sandbox`	no	second process for networked Play-in-Editor
`Saved/`	created	logs, crash reports, caches and editor settings
`Saved/Config/editor.ini`	created	panel layout and editor settings for this machine
`build-dev/bin/plugins/`	no	native engine plugins
Vulkan Loader installed on the system	yes	window and GPU rendering on Windows and Linux
`Projects/`	no	default location for new projects

Selecting a project

A project is a directory with a `project.femproj` file and `assets/`, `configs/` subdirectories. New projects are created in `Projects/` inside the engine root; an existing one is opened from the editor dashboard by the path to `project.femproj` or to the project folder. There is no command-line argument for the project: the working directory decides, which is why a "project shortcut" is a shortcut with a different working directory.

Development tools

Besides the editor, `build-dev/bin` contains separate executables. All of them are console programs that read files and need no GUI.

Tool	Purpose	Files it needs
`fby_editor`	editor: scenes, assets, PIE, text preview	`assets/`, `project.femproj`
`fby_sandbox`	sandbox and Play-in-Editor client	`assets/`
`fby_server`	dedicated server	project directory if configured
`fby_lobby`	lobby and matchmaking service	none
`fby_fontc`	build a `.fbyfont` from a TTF/OTF	the input TTF
`fby_textc`	compile marked-up text into `.fbin`	the input `.toml` text
`fby_fempak`	pack assets into a `.fempak`	the asset directory
`fby_scene_tool`	validate and convert scenes	the scene `.toml`
`fby_template_gen`	generate a new project skeleton	destination directory
`fby_aotc`	nativize scripts (AOT)	project with bytecode
`fby_loc_tool`	validate and build localization tables	localization `.toml`
`fby_shaderc`	compile shaders to SPIR-V	`.vert`/`.frag`; the build calls it automatically

The tools live in `build-dev/bin/tools`; the test executables in `build-dev/bin/tests`.

Build and test through the CMake presets; dependency setup lives in `SetupScripts/*`.

Shortcuts for fast launching

The working directory is the only thing that distinguishes "launching the engine" from "launching a project", so a shortcut must always set it explicitly.

Windows

25. Right-click `build-dev\bin\fby_editor.exe` → Send to → Desktop (create shortcut).
26. Open the shortcut properties: "Start in" = project root (for example `%USERPROFILE%\FemboyEngine`), "Target" = full path to `fby_editor.exe`.
27. Rename it to "FemboyEngine" and drag it to the taskbar or the Start menu.
28. For a specific project, create a second shortcut with the same target and "Start in" = the project folder. That is the project selection without a dialog.

macOS

29. Open the project folder in Finder.
30. Drag `fby_editor` (or the `.app` bundle, if built) to the Dock while holding `Command`.
31. For a specific project create a `.command` file with two lines: `cd /path/to/project` and `/path/to/fby_editor`; Finder runs it on double-click and the alias can live in the Dock.
32. `SetupScripts/Setup.command` starts the same way — by double-click.

Linux

Create `~/.local/share/applications/fby-editor.desktop`:

```
[Desktop Entry]
Type=Application
Name=FemboyEngine
Exec=/path/to/fby_editor %f
Path=/path/to/project
Icon=/path/to/Logo.png
Terminal=false
Categories=Development;
```

Then pin it to the dock or the panel. A second `.desktop` file with a different `Path` gives a shortcut to another project.

Android and iOS

Mobile builds use the SDL3 platform layer: window, input and the application lifecycle come from SDL; rendering is Vulkan on Android and Metal (or Vulkan through MoltenVK) on iOS. The editor does not build for mobile platforms.

Android: Android Studio, NDK `28.2.13676358` (r28c), OpenJDK 17, Gradle 8.12, Android Gradle Plugin 8.7.3, `compileSdk 35`, minimum API level 21. CMake configures Android through the NDK toolchain file; packaging uses the SDL3 Gradle project.

iOS: macOS with Xcode 16 or newer, iOS SDK 14.2+, minimum deployment target iOS 11.0. CMake configures with `-DCMAKE_SYSTEM_NAME=iOS`; signing and packaging use Xcode.

The exact versions can always be checked with:

```
cmake --version
ninja --version
```

and, for mobile, `ls "$ANDROID_NDK_HOME/source.properties"` and `xcodebuild -version`.

Troubleshooting

Symptom	Cause	Fix
The editor shows no assets	it was started outside the project root	start it from the directory with `assets/` and `project.femproj`
Vulkan error at startup	the system has no Vulkan Loader	install the Vulkan Runtime or build with `-DFBY_RHI_BACKEND=null`

The editor or a game closed without a message	the process crashed	open `Saved/Crashes/` in the working directory: `report.txt` and `crash.dmp`
A run log is needed for a report	logs are written to a file now	take the newest file from `Saved/Logs/<category>/`
`glslang` missing during the build	`third_party/glslang-src` is missing	run the setup script again
No fonts in the editor	the TTFs or the family manifest are missing	restore `assets/fonts/` from the repository
CMake reports a missing harfbuzz or SheenBidi	the `third_party/` directory was damaged or removed	restore `third_party/harfbuzz-download/` and `third_party/sheenbidi-download/` from the repository
The Android build cannot find the NDK	`ANDROID_NDK_HOME` is unset	point it at NDK r28c and reconfigure
The Windows build fails on headers	the Windows SDK is missing	install Build Tools with the C++ workload